Image Processing 31651

Micro-Project 2: Usage of Filtration

(30 Points to the Grade)

No PHYSICALL presentation

Report must be emailed before

2024-05-16 07:59:59:000

Preparing MP2

When MP2 is finished, students are expected to do the following:

1. **Finalize** Report as PowerPoint file and save it as PDF file: **ABCD-EFGH-IJKL-MP2.pdf**
2. Create zip file with the name **ABCD-EFGH-IJKL-MP2.zip** **(ZIP only – no RAR or Z7)**
3. Add to **ABCD-EFGH-MP2.zip** relevant to the specific tasks folders containing files *.h, *.cpp, *.sln, *.vcxproj
5. If relevant, add to relevant subfolders of **ABCD-EFGH-IJKL-MP2.zip** source images in the BMP format (images that were used to create processed images).

It is forbidden to add images that are to be created by student' code in the frames of the specific assignment.

6. Email files **ABCD-EFGH-IJKL-MP2.pdf and ABCD-EFGH-IJKL-MP2.zip** to the Email of the second and to the third student.

It is a duty of the second and the third student
to download those files,
unzip zip,
open relevant subfolder (one by one)
double press to *.sln file,
compile the project
run created executable(s)
validate that all images mentioned in the **ABCD-EFGH-IJKL-MP2.pdf**
were created with the same names as requested in this report

Sending MP2

Send MP2 files not later than 2024-05-16 07:59:59:000
to ssk.improc@gmail.com

SUBJECT **"ABCD-EFGH-IJKL-MP2"**
***Text of the Email*:** **Name of the presentation,**
Shorten Private Names of 3 students
***Attachment*:**
ABCD-EFGH-IJKL-MP2.pdf
zip file (zip, no rar, no Z7) having name: ABCD-EFGH-IJKL-MP2.zip
as described in the slide Preparing MP2

Create Test Image and blur it by using Convolution

1. (2 points) By using MS Paint and BMP Library (or any other software)
   create Gray BMP VGA Image (640x480) Tim1.bmp containing 6 lines of the text:
   "Hello from ABCD-EFGH-IJKL" (Replace RED)
   * Font size of every line is differ (from small to large)
2. (4 points) By using convolution with **Fast** Gaussian filter 5x5 and 7x7, BLUR Image:

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1	4	7	4	1
4	16	26	16	4
7	26	41	26	7
4	16	26	16	4
1	4	7	4	1

1/1003

0	0	1	2	1	0	0
0	3	13	22	13	3	0
1	13	59	97	59	13	1
2	22	97	159	97	22	2
1	13	59	97	59	13	1
0	3	13	22	13	3	0
0	0	1	2	1	0	0

Resulted images: Tim1LPF5c.bmp and Tim1LPF7c.bmp

Select font sizes so that after the blurring at least 2 lines will be hardly readable and 2 lines unreadable

Blur test Image by using FFT and “restore it”

3 (2 points) Use filtration with FFT to blur Tim1.bmp by using Gaussian Filter in FD.
Select size of the Gaussian Filter so that at least 4 lines will be hardly readable and unreadable.

Resulted image: Tim1LPFF.bmp

4 (6 points) In attempt to “restore” above blurred images (by using HPF convolution filters), find at least two HPF and apply them to above blurred images. Add to “restored images” prefix HPF1c and HPF2c

5 (6 points) In attempt to “restore” above blurred images by using FFT, “invent” HPF filter in FD and apply it to above blurred images. Add to “restored images” prefix HPFf

6 (4 points) Properly number all images and profiles in the report and explain your results by referring all the images: what was good and wat was not good.

7 (6 points) Repeat above steps for the grayImage13bmp from Assignment 13 (TV Test Image).
To retain naming scheme, rename it to **Tim2.bmp**

*The code for the convolution can use static allocation of the Images,
but pixels must be accessed by using pointer and not as Image[row][column][color]
*Using OOP is not a must
*Students are expected to understand when to use unsigned char, integer or float/double
* In case more filters were tried, document your attempts. Your reasonable attempts will be valued